BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: BN-IX-XXIII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-B)

Time Allowed: 20 Min | **Maximum Marks:** 15 | **Paper Code:** BN-IX-XXIII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If z = 2 - 9i then conjugate of z is:

(A) -2 + 9i
(B) 2 + 9i
(C) 9i
(D) -2

Q2. $8.3x10^{-5}$ can be written in standard form as:

(A) 0.000083
(B) 0.00083
(C) 0.0000083
(D) 0.8300000

Q3. Which of the following is the base of common logarithm?

(A) π
(B) e
(C) 10
(D) 5

Q4. Which of the following is a polynomial?

(A) $y^2 + 1/y^2 + 2$ (where $y \neq 0$)
(B) $y^2 + y^{-3} + 3$
(C) $y^2 + y^{-4} + 4$
(D) $y^2 + y^{-5} + 5$

Q5. $(2p - q)^3 =$

(A) $8p^3 - q^3 + 6pq\ (2p - q)$
(B) $8p^3 + q^3 + 6pq\ (2p + q)$
(C) $8p^3 + q^3 - 6pq\ (2p + q)$
(D) $8p^3 - q^3 - 6pq\ (2p - q)$

Q6. The factorization $4x^2 + 4xy + y^2$ is

(A) (2x - y)(2x - y)
(B) (2x + y)(2x + y)
(C) 4x (x + y) + y^2
(D) 4x (x + y) - y^2

Q7. If C and D are two polynomials and their HCF and LCM are respectively H and L then D =

(A) C / (H×L)
(B) (H×L) / C
(C) C / L
(D) C / H

Q8. The solution of linear equation 2z + 3 = -6z + 5 is:

(A) 1/4
(B) -1/4
(C) 2
(D) -2

Q9. If S = {3,7} and T = {1,2}, then which of the following is ordered pair of S x T?

(A) {(3,1), (1,2)}
(B) {(7,1)}
(C) {(1,3)}
(D) {(2,7), (1,3)}

Q10. Which of the following elements represent one of the columns of the matrix [[7, 2, 5], [4, 9, 8],[1, 6, 3]]

(A) 7, 2, 5
(B) 4, 9, 8
(C) 7, 9, 3
(D) 2, 9, 6

Q11. Which of the following two matrices are equal? D = [[3-4, 2+3],[-6, 0+4]], E = [[4-5, 1+4],[-12+6, 4]], F = [[3-5, 3+3],[1, 0]], H = [[6-6, 0+5],[1, 0]]

(A) D and E
(B) E and F
(C) F and H
(D) H and D

Q12. Which of the following is an identity matrix?

(A) [[0, 0, 0],[0, 0, 0],[0, 0, 0]]
(B) [[1, 1, 1],[1, 1, 1],[1, 1, 1]]
(C) [[1, 0, 0],[0, 1, 0],[0, 0, 1]]
(D) [[1, 1, 0],[0, 1, 1],[1, 1, 0]]

Q13. If R== [[-2, 3],[4, -5]] and T= [[1, -3],[-1, -4]] the R-T is equal to:

(A) [[-1, 0],[3, -1]]
(B) [[-3, 6],[5, -1]]
(C) [[3, 0],[3, 1]]
(D) [[-2, -9],[-4, 20]]

Q14. The rational number 4/3 can be expressed as:

(A) 4.666666667
(B) 0.2142857143
(C) 1.33333333
(D) 0.33333333

Q15. y^(2/3) can be expressed in radical form as:

(A) $\sqrt[7]{y^3}$
(B) $\sqrt[7]{y}$
(C) $\sqrt[3]{y^7}$
(D) $\sqrt[3]{y}$

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36 | **Paper Code:** BN-IX-XXIII-I

Note: Attempt section B & C accordingly.

Q2. (Marks 36) Q.No 2. Attempt any NINE parts of the following. All parts carry equal marks.

i. If U = [[4, -9],[4, 11]], V = [[10, 7],[4, 3]] determine whether UV = VU, or not?

ii. Simplify: (a) $(6\ c^2\ d^3)^3$ (b) $(-8\ u^3\ v)^2$

iii. Find the value of 'q': $\log\sqrt{2}\ 49 - \log\sqrt{2}\ 7 + \log\sqrt{2}\ 3 = \log\sqrt{2}\ q$

iv. Evaluate the following when t = -2: $(2t^2 - 5t + 2) / (t - 3)$

v. Find the value of $p^2 + q^2$ if p + q = -8, p - q = -4.

vi. Without performing division, find the remainder when $3b^3 - 4b^2 + 2b - 1$ is divided by b + 1.

vii. Simplify: $3/(y+2) - 2/(y+2) - 4/(y^2+2)$

viii. Solve $\sqrt{(3x + 5)} + 5 = 7$

ix. Solve and plot on number line: $5x + 4 > 34$, $x \in N$.

x. Prove that the points A(1,-1), B(5,2) and C(9,5) are collinear.

xi. Prove that: if two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

xii. Prove that any point on the right bisector of a line segment at equidistant from end points of the segment.

SECTION – C

Maximum Marks: 24 | **Paper Code:** BN-IX-XXIII-I

Note: Attempt any FOUR of the following. All questions carry equal marks.

Q3. Q.No:3. Prove that if two opposite sides of a quadrilateral are congruent, then it is a parallelogram.

Q4. Q.No:4. Prove that from a point outside a line, the perpendicular is the shortest distance from the point to the line.

Q5. Q.No:5. Prove that if a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side.

Q6. Q.No:6. Prove that in a right-angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

Q7. Q.No:7. Prove that parallelograms on equal bases and having the same altitude are equal in area.

Q8. Q.No:8. Construct ΔQRS, for $mQR = 8cm$, $mRS = 12cm$, and $m\angle R = 90°$.